

AML — Presentation & Diagnosis



1. Pale soldier (anemia)

WHAT YOU SEE

A pale, motionless soldier asleep on the ground in the bone-marrow-failure zone — the drained figure is the normocytic anemia from blasts crowding out red-cell production.

WHAT IT ENCODES

Blasts replace normal marrow, crowding out erythropoiesis and producing a normocytic, normochromic anemia. Patients present with fatigue, pallor, exertional dyspnea, and sometimes new/worsening angina or heart failure. Anemia plus thrombocytopenia plus neutropenia (pancytopenia) with circulating blasts is the classic AML picture, and the dominant complaint is often simply profound fatigue.

BOARD TRAP

AML anemia is normocytic — a macrocytic anemia should redirect you toward megaloblastic causes or MDS, and a microcytic one toward iron deficiency. Pancytopenia with blasts on smear is AML until proven otherwise.

2. Bruises / petechiae

WHAT YOU SEE

A soldier mottled with purple glaze spots and bruises — the purple skin blotches are petechiae/ecchymoses from marrow-failure thrombocytopenia.

WHAT IT ENCODES

Thrombocytopenia from marrow failure causes mucocutaneous bleeding: petechiae, ecchymoses, gum bleeding, epistaxis, and menorrhagia. In acute promyelocytic leukemia (APL, t(15;17)/M3), bleeding is far worse because of overt DIC — the promyelocyte granules release tissue factor, consuming clotting factors and platelets and causing life-threatening hemorrhage (often intracranial).

BOARD TRAP

Severe bleeding/DIC at presentation should make you think APL (M3) — start ATRA immediately on clinical suspicion before genetic confirmation, because untreated APL coagulopathy is a leading cause of early death. Do not wait for cytogenetics.

3. Microorganism (infection)

WHAT YOU SEE

A microbe symbol beside a collapsed soldier — the lurking bug is opportunistic infection from functional neutropenia despite a high WBC.

WHAT IT ENCODES

Despite a high total white count, the circulating cells are non-functional blasts, so patients are functionally neutropenic and present with recurrent, severe, or opportunistic infections and neutropenic fever. Neutropenic fever (ANC <500 with temp ≥38.3°C once or ≥38.0°C sustained) is an emergency requiring empiric broad-spectrum antibiotics within an hour.

BOARD TRAP

A high total WBC in AML does NOT mean functioning neutrophils — the patient is effectively neutropenic and immunocompromised. Never reassure based on the total WBC; assess the absolute neutrophil count and treat fever as an emergency.

4. Massively swollen gums

WHAT YOU SEE

A patient in the triage tent with grotesquely swollen, overgrown gums and nodules — the bulging gums are gingival infiltration of monocytic AML (M4/M5).

WHAT IT ENCODES

Gingival hypertrophy/infiltration and leukemia cutis (skin nodules/plaques) are hallmarks of monocytic differentiation — AML M4 (myelomonocytic) and especially M5 (monoblastic/monocytic). Monocytic AML also carries higher rates of CNS involvement and leukostasis. Monocytic markers include CD14, CD11b, CD64 and nonspecific esterase positivity.

BOARD TRAP

Gum infiltration and leukemia cutis point specifically to MONOCYTIC AML (M4/M5), not leukemia in general. Tying these physical findings to the monocytic subtype is the testable association.

5. WBC meter >100k

WHAT YOU SEE

A WBC meter on the board pinned hard into the red past 100k — the maxed-out dial is hyperleukocytosis causing leukostasis.

WHAT IT ENCODES

Hyperleukocytosis (WBC, particularly blast count, >50,000-100,000/μL) makes blood viscous and lets rigid blasts plug the microvasculature — leukostasis. Risk is highest in monocytic subtypes. Management is urgent cytoreduction: hydroxyurea and/or leukapheresis as a bridge, with prompt definitive induction; avoid RBC transfusion before cytoreduction as it further raises viscosity.

BOARD TRAP

Leukapheresis is a temporizing emergency measure for symptomatic leukostasis, not definitive treatment — start induction chemo promptly. Avoid packed-RBC transfusion in hyperleukocytosis until the count is reduced, since it worsens hyperviscosity.

6. Brain / lung crash

WHAT YOU SEE

Brain and lung icons toppling off the edge of the meter board — the crashing organs are leukostasis hitting CNS and lungs.

WHAT IT ENCODES

Leukostasis manifests as CNS dysfunction (headache, confusion, visual changes, stroke, intracranial hemorrhage) and pulmonary failure (dyspnea, hypoxia, diffuse infiltrates). It is a medical emergency with high early mortality; suspect it in any hyperleukocytotic AML patient with neurologic or respiratory deterioration and cytoreduce urgently.

BOARD TRAP

New confusion or hypoxia in a patient with WBC >100k is leukostasis, not pneumonia or stroke alone — the treatment is emergent cytoreduction (leukapheresis/hydroxyurea + chemo), not just supportive care.

7. Auer rod

WHAT YOU SEE

A pink needle-shaped Auer rod symbol tagged NEVER ALL on the meter board — the crystalline needle is pathognomonic for AML and never seen in ALL.

WHAT IT ENCODES

Auer rods are pink/red needle-like crystalline aggregates of fused azurophilic (primary) granules and MPO seen in the cytoplasm of myeloblasts — they are pathognomonic for AML and never occur in ALL. Bundles of multiple Auer rods (faggot cells) are characteristic of APL (M3). On stains, AML blasts are MPO-positive and Sudan-black-positive.

BOARD TRAP

Auer rods exclude ALL — they confirm myeloid lineage. Faggot cells (multiple Auer rods bundled) specifically suggest APL/M3 and should trigger immediate ATRA. ALL blasts are MPO-negative and TdT/PAS-positive instead.

8. PLT meter <100k

WHAT YOU SEE

A platelet meter pinned low in the red below 100k — the bottomed-out dial is the thrombocytopenia driving the bleeding picture.

WHAT IT ENCODES

Thrombocytopenia results from marrow replacement by blasts and is one component of the presenting pancytopenia. It drives the bleeding manifestations and is far more dangerous when superimposed on the DIC of APL. Platelet transfusion thresholds are higher when there is active bleeding, fever, or APL coagulopathy.

BOARD TRAP

Thrombocytopenia from marrow failure plus DIC in APL is a deadly combination — keep platelets and fibrinogen well supported in suspected APL while ATRA is started, rather than using the usual <10k transfusion threshold.

9. Hb low (normocytic)

WHAT YOU SEE

An Hb meter reading low beside normocytic-normochromic cell circles — the dial plus normal-sized cells is the normocytic anemia of marrow infiltration.

WHAT IT ENCODES

The anemia of AML is normocytic and normochromic, reflecting decreased production from marrow infiltration rather than a maturation or iron defect. The blood smear shows circulating blasts alongside the cytopenias, and the diagnosis is confirmed on bone marrow with cytogenetics, flow cytometry, and molecular studies.

BOARD TRAP

Normocytic normochromic anemia is the rule in AML — a macrocytic picture should prompt thought of MDS/B12-folate deficiency, reminding you that MDS can transform into AML over time.

10. Chloroma / sarcoma mass

WHAT YOU SEE

A glowing green mass erupting on the soldier's spine/legs — the green tumor is a myeloid sarcoma (chloroma), itself AML-defining.

WHAT IT ENCODES

A myeloid sarcoma (chloroma, granulocytic sarcoma) is an extramedullary tumor of myeloblasts — green on cut surface from MPO. It can arise in skin, bone, lymph nodes, orbit, or CNS and may precede or occur without overt marrow involvement. Regardless of marrow blast percentage, an extramedullary myeloid sarcoma is itself diagnostic of AML and treated as systemic disease.

BOARD TRAP

A myeloid sarcoma diagnoses AML even when the marrow has <20% blasts — extramedullary tissue disease counts as AML and warrants full systemic therapy, not local treatment alone.

11. Back/bone pain

WHAT YOU SEE

A soldier clutching his back with the spine lit up — the gripped, glowing spine is bone pain from marrow expansion under blast pressure.

WHAT IT ENCODES

Bone and joint pain (often sternal tenderness or back pain) arises from marrow expansion under pressure from proliferating blasts. It is more prominent in children and can mimic other causes of bone pain. Hepatosplenomegaly and lymphadenopathy are generally less prominent in AML than in ALL or chronic leukemias.

BOARD TRAP

Bone pain plus cytopenias in a child can mimic juvenile arthritis or osteomyelitis — a peripheral smear and marrow exam distinguish leukemia. Marked lymphadenopathy/organomegaly favors ALL or CLL over AML.

12. Age >3 months / acute

WHAT YOU SEE

A short timeline trench marked in months on the ridgeline — the brief timeline is AML's acute days-to-weeks onset, unlike chronic leukemias.

WHAT IT ENCODES

AML is an ACUTE leukemia: symptoms evolve over days to a few weeks from rapidly accumulating blasts, in contrast to the indolent course of chronic leukemias. AML is the most common acute leukemia in adults and its incidence rises sharply with age; antecedent MDS or prior chemo shortens the prodrome and worsens prognosis.

BOARD TRAP

A slow, indolent course over months to years argues against acute leukemia and toward a chronic leukemia or MDS. AML predominates in older adults, whereas ALL is the most common acute leukemia of childhood.